JLU-SPH

ByeGerm 1.0 User Manual

Designed for small-scale testing at home.



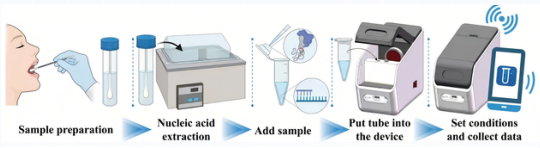


Contact us!

Before you begin

Prepare the supplied power source, an Android phone with the matching app, labeled collection and reaction tubes, sampling materials, reagents, pipetting equipment, controls, protective equipment, and waste containers. Use the reagent protocol for all volumes, temperatures, times, and result criteria.

Testing Workflow



1 Collect and label the sample

Have a trained healthcare professional collect the throat swab following the sampling kit instructions. Seal the collection tube, record the sample ID and collection time, and check for leakage. Keep the same ID through processing and testing.

2 Prepare the sample

Perform sample lysis or nucleic acid extraction outside the instrument, following the test protocol. Use suitable external heating equipment where required. Keep collection tubes out of the detection positions.

3 Prepare the reactions

Add processed sample to the reaction mixture according to the reagent protocol. Include the required controls, mix as instructed, and close all caps. Keep labels clear of the imaging area.

4 Load the tubes and start the run

Place the instrument on a dry, stable bench with clear vents. Check the cable and loading area, use the supplied power source, and wait for the ready indication. Before first use or after repairs, have the responsible person verify heating, imaging, saving, and stopping functions.



Connect



Set temperature and time



Monitor the run

App examples Use the parameters specified by your test protocol

Connect the phone

Join the instrument Wi-Fi network. Enter its configured Server IP and Port in the app and tap Connect to Server. Confirm that status and temperature readings update.

Load and record positions

With the instrument stopped and ready for loading, place each sealed reaction tube in its assigned fixed position. Record the run ID, sample ID, target, and tube position, including controls. Seat tubes securely and close the lid.

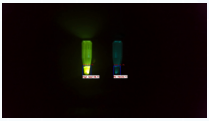
Set and start

In Settings, enter the protocol temperature and run time. Check controls, tube positions, closed lids, and the connection, then start. Confirm that Current Temp and Heating countdown update. Record the start time using the protocol timing reference.



5 View and save the results





Actual fluorescence image

Example results screen and captured tube image

Capture and check

Keep the phone connected and the lids closed during the run. Capture images at the protocol acquisition stage using the configured automatic or manual mode. Check image clarity, timestamps, and tube positions against the loading record.

Check controls and run records before interpreting results under the reagent protocol. Blank or old data are not negative results. Brightness, model confidence, and app risk labels alone do not establish a test conclusion.

Save the record

Save or capture the results screen. Retain the run ID, sample-to-position mapping, run parameters, results, reagent lot, and protocol version. Mark unused positions as unused.

Stop and clean

Confirm that heating has stopped and allow cooling before removing tubes. Keep caps closed, account for all tubes, and dispose of waste under institutional rules. Shut down, disconnect power, and clean with a compatible method. Keep liquid out of optics, vents, and connectors.

If something goes wrong

Connection lost: check the instrument state, reconnect, and verify run and data completeness. Closing the app does not stop heating.

Missing images or failed controls: follow the protocol for handling or retesting. Liquid ingress or abnormal heating: stop use and disconnect power when safe; contact the responsible person.

